

InfoSec PreMid Quiz Name: _____ Roll No: _____ Date: April 1st, 2026
Marks: 30

1. If an adversary is passive and only performs eavesdropping, which pillar of the CIA triad is most likely compromised?

- A. Confidentiality
- B. Availability
- C. Integrity
- D. Reliability

2. Symmetric encryption is also known as:

- A. Public-key encryption
- B. Private-key encryption
- C. Shared-key encryption
- D. Secret-key encryption

3. Ciphertext: IKEWGPEQPVTHPXIOHMSLB, Key: PLAYFAIR, Cipher name is Playfair. Plaintext is

- A. HIDETHESILVERINTHEFOREST
- B. HIDETHEGOLDINTHEFOREST
- C. FINDTHEGOLDINTHECAVE
- D. BURYTHEGOLDINTHEFOREST

4. Which type of system monitors network traffic in real-time to look for patterns or signatures of known attacks?

- A. HIDS (Host-based IDS)
- B. NIDS (Network-based IDS)
- C. Firewall
- D. Wireshark

5. Which of the following can prevent a MITM attack in Diffie-Hellman?

- A. Using larger prime (p)
- B. Using XOR instead of mod operations
- C. Authenticating public keys with digital signatures
- D. Using symmetric encryption

6. Hashing is designed to be a one-way function. This means:

- A. It can only be used by one person
- B. The original input cannot be derived from the output
- C. It only works on the Physical layer
- D. It requires a public and private key

7. If a message is encrypted using Alice's public key, who can decrypt it?

- A. Anyone with Alice's public key
- B. Only Alice
- C. Only Bob with private key
- D. Anyone with Bob's private key

8. Alice and Bob use Diffie-Hellman and later encrypt messages with AES using the derived key. Oscar records all messages but never interferes. Later, Oscar steals Bob's private key.

What can Oscar do now?

- A. Decrypt all past messages
- B. Nothing
- C. Decrypt only future messages
- D. Both past and future messages

9. Before encrypting with Playfair, the message BALLOON is prepared into digraph pairs. What is the correct prepared form?

- A. BA LL OO NX
- B. BA LX LO OX NX
- C. BA LX LO ON
- D. BA LO ON XX

10. Which countermeasure is directly linked with maintaining integrity of data?

- A. Data replication
- B. Hashing
- C. Regular backup only
- D. Adequate bandwidth

11. Which of the following is NOT a modern symmetric cipher?

- A. AES
- B. RSA
- C. Blowfish
- D. RC4

12. Which OSI layer includes Cutting cables and Jamming as attacks?

- A. Physical Layer
- B. Network Layer
- C. Data Link Layer
- D. Transport Layer

13. Which Wireshark display filter finds HTTP traffic specifically containing the string "example.com"?

- A. http.request
- B. dns.query == "example.com"
- C. http.host contains "example.com"
- D. ip.dst == example.com

14. To identify a potential "SYN flood" attack in Wireshark, which filter is most useful?

- A. tls.handshake.type == 1
- B. port 53
- C. http.host contains "example.com"
- D. tcp.flags.syn == 1

15. In the digital signature verification process, the recipient uses the _____ key to verify the signature.

- A. The recipient's public key
- B. The sender's private key
- C. The recipient's private key
- D. The sender's public key

Short Questions

1. What is meant by zero-day exploits.

A zero-day exploit is an attack that targets a software vulnerability previously unknown to the vendor or the public. The name refers to the fact that developers have had "zero days" to fix the issue.

2. Difference between Addressing and Routing?

Addressing is the process of identifying a unique device or destination in a network (the "Where"). Routing is the process of determining the best path for data to travel to reach that destination (the "How").

3. What role do certificates play in a TLS handshake?

Certificates provide authentication by proving the server's identity. They prevent Man-in-the-Middle attacks by ensuring the client is communicating with the legitimate owner of the public key.

4. Difference between Symmetric Encryption and Asymmetric Encryption.

Symmetric encryption uses a single shared secret key for both encryption and decryption. Asymmetric encryption uses a pair of keys: a public key (for encryption) and a private key (for decryption).

5. **Your company wants to block all incoming traffic from the internet except web browsing (HTTP/HTTPS) to its public web server. What basic firewall rule would you apply?**

```
iptables -A INPUT -p tcp -m multiport --dports 80,443 -j ACCEPT    (Allow web traffic)
iptables -P INPUT DROP    (Block everything else)
```

Long Question:

1. **What is Diffie-Hellman protocol? Explain its usage and what problem it solves? Did you find any vulnerability in it, please explain.**

Diffie-Hellman (DH) Protocol

Developed by Whitfield Diffie and Martin Hellman in 1976, it is a mathematical method that allows two parties (Alice and Bob) to establish a shared secret key over an insecure communication channel.

Working:

Both parties agree on public parameters: a large prime number (p) and a primitive root (g). Each party chooses a **private** secret number. They exchange **public** values calculated using modular exponentiation.

By performing further math on the received public values, both Alice and Bob arrive at the **exact same** secret key.

What problem does it solve?

It solves the **Key Distribution Problem**. In symmetric cryptography (like AES), both parties need the *same* secret key to communicate. Historically, getting that key from Alice to Bob was dangerous because an attacker could intercept it during transit.

The DH Solution: Alice and Bob never actually send the secret key to each other. They only exchange public information that allows them to compute the key independently at their respective ends.

Vulnerabilities in Diffie-Hellman

The primary vulnerability in the basic Diffie-Hellman protocol is the **Man-in-the-Middle**.

Basic DH is **unauthenticated**. Alice knows she received a public value, but she has no way to prove it actually came from Bob.

The Scenario (Referencing "Darth" in your slides):

1. Darth sits between Alice and Bob.
2. When Alice sends her public value , Darth intercepts it and sends Bob his own public value instead
3. Bob does the same, and Darth intercepts Bob's value too.
4. Now, Darth has established two separate secret keys: one with Alice and one with Bob.
5. When Alice sends an encrypted message, Darth decrypts it with the first key, reads it, re-encrypts it with the second key, and sends it to Bob. Alice and Bob believe they are speaking privately, but Darth is controlling the entire conversation.

2. Differentiate between Display and Capture Filter of wireshark, give 3 examples of each.

Capture Filters: Applied before the capture starts. Once a capture starts, you cannot change the filter without restarting. High efficiency; saves CPU and disk space by ignoring unwanted traffic.

Examples of Capture Filters

- | | |
|---------------------|--------------------------------------|
| 1. host 192.168.1.1 | Capture only traffic to/from this IP |
| 2. port 80 | Capture only HTTP traffic |
| 3. tcp and port 443 | Capture only HTTPS TCP traffic |

Display Filters: Applied after the capture is finished or while it is running. Can be changed or removed at any time without losing data. Lower efficiency; all traffic is processed, but only specific packets are visible.

Examples of Display Filters

- | | |
|---------------------------|---------------------------------|
| 1. ip.addr == 192.168.1.1 | Show packets with this IP |
| 2. http | Show only HTTP protocol packets |
| 3. tcp.port == 443 | Show only HTTPS traffic |